

Consider a sequence

$$x = \{x_1, x_2, \dots, x_T\}$$

To incorporate the ordinality of the input sequence, another fixed "temporal variable" needs to be added.

$$t = \{0, 1, 2, \dots, 3\}$$

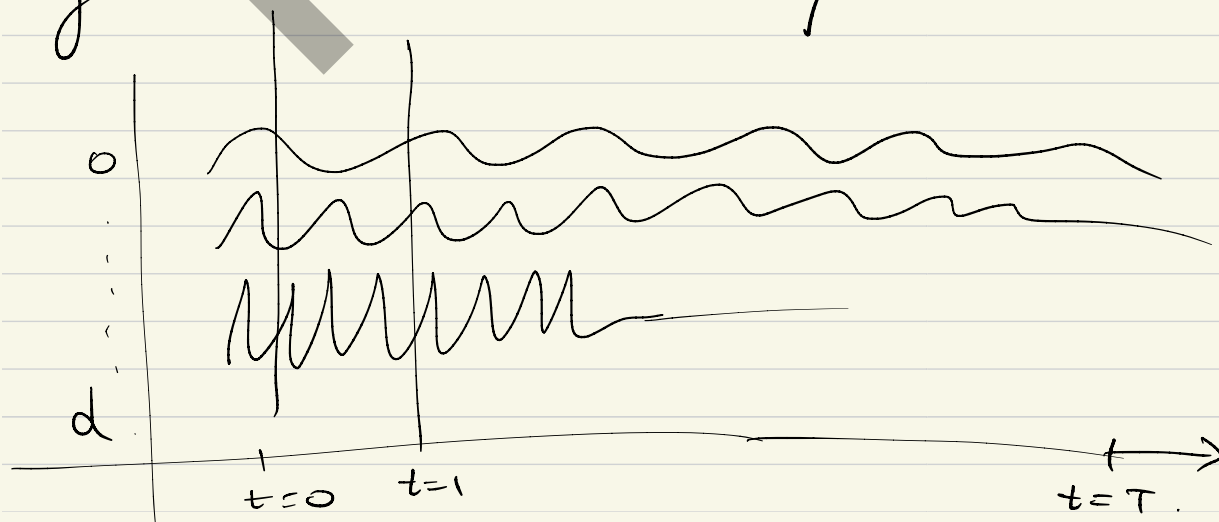
$$\hat{x}_i = x_i + t_i$$

scalar t gets ignored & $\therefore$ a fixed vector of dim d needs to be considered.

$$f: t \rightarrow \hat{t}, \quad \mathbb{R} \rightarrow \mathbb{R}^d$$

$\hookrightarrow$ positional Embedding.

Eg: Sinusoidal Embedding



$$\hat{t}(i, 2t) = \sin\left(\frac{i}{N^{2t/d}}\right)$$

$$\hat{t}(i, 2t+1) = \cos\left(\frac{i}{N^{2t/d}}\right)$$

$$\hat{x}_i = x_i + \hat{t}_i$$

Transfer Learning

Given $x \in \mathbb{R}^d$, $y \in \mathbb{R}$,

suppose $g_\phi(x): x \rightarrow y$

$g_{\phi^*}()$ is a NN, learned via ERM.

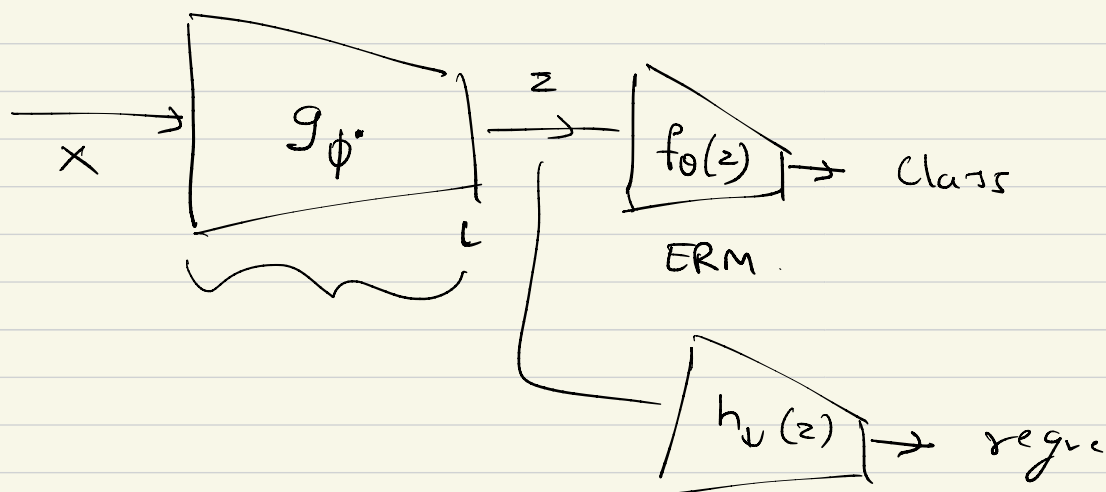
↳ "pre-trained NN"

Since $g_\phi()$ is a composite function,

the outputs can be tapped from any layer

$z = g_{\phi^*}(x) \big|_{\text{layer}=l}$, Embedding of x
from g_{ϕ^*}

Use z for any further down-stream tasks.



multi-task learning

Knowledge distillation.

Suppose $x, y \sim \mathcal{P}_{xy}$.

$g_\phi()$ is trained on a task.

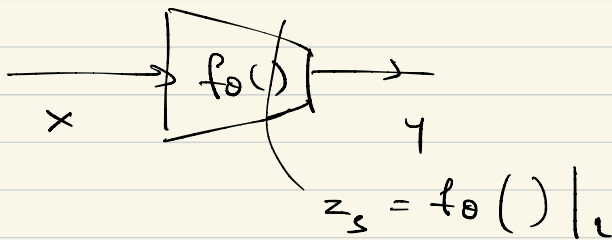
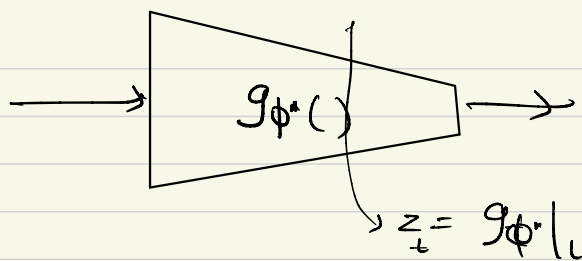
↳ teacher Network

Let $f_\theta()$ be another NN (typically

↳ student network.

with lesser parameters than g_ϕ)

question: How to get the performance of the student network to match that of the teacher?



Ensure that $\dim(z_t) = \dim(z_s)$

Regularize the student network s.t

$D_r(p_{z_t} \parallel p_{z_s})$ is minimized.

$$\hat{R}_{\text{student}}(\theta) = \operatorname{argmin}_{\theta} \left[D(p_{z_t} \parallel p_{z_s}) + \frac{1}{N} \sum \ell(f_{\theta}(), y) \right]$$

Training NNs with Adaptive learning rate

NNs are trained with ERM, via gradient descent.

suppose $\hat{R}(\theta)$ is the ER, grad descent

$$\theta^{t+1} \leftarrow \theta^t - \alpha \nabla_{\theta} \hat{R}(\theta)$$

↳ fixed learning rate

Stochastic gradient descent

$$\hat{R}(\theta) = \frac{1}{B} \sum_B \ell(h_\theta(x), y)$$

where $B \subset D$, randomly sampled.

Adaptive learning rate for grad-des

i) SGD with Momentum

$$\text{Define } m_t = \beta_1 m_{t-1} + (1 - \beta_1) g_t$$

$$\text{where } g_t = \nabla_\theta \hat{R}(\theta)$$

$$\theta^{t+1} \leftarrow \theta_t - \alpha m_t$$

2) RMS prop :

$$v_t = \beta_2 v_{t-1} + (1 - \beta_2) g_t^2$$

$$\theta^{t+1} \leftarrow \theta_t - \frac{\alpha}{\sqrt{v_t}} \odot g_t$$

3) Adaptive momentum Estimation (Adam)

$$m_t = \beta_1 m_{t-1} + (1 - \beta_1) g_t$$

$$v_t = \beta_2 v_{t-1} + (1 - \beta_2) g_t^2$$

$$\theta^{t+1} \leftarrow \theta_t - \frac{\alpha}{\sqrt{v_t}} \cdot m_t$$